RURALCRAFT GLOBAL

A Unified Cross-Border Rural Handicraft Marketplace with Automated Customs Documentation & HS Code Classification Engine

EXECUTIVE OVERVIEW

RuralCraft Global is an enterprise-grade, direct-to-consumer (D2C) cross-border e-commerce platform designed to bridge Indian rural artisans directly with international buyers. It integrates a multi-currency conversion engine, automated Harmonized System (HS) export tariff classification, instant customs commercial invoice PDF generation, and a unified command center managing customer orders, artisan profiles, and verified export logistics.

PROJECT CREDENTIALS & METADATA

Project Title:	RuralCraft Global — Cross-Border Artisan Platform
Submitted By:	Manoj B (Student / Developer)
Domain:	Full-Stack E-Commerce & Customs Tech
Core Tech Stack:	React 19, TypeScript, Supabase, Tailwind, jsPDF
Deployment Target:	Vercel (Single Unified SPA Deployment)
Database Cloud:	Supabase Cloud (PostgreSQL with RLS)
Academic Year:	2025 – 2026

1. ABSTRACT & PROJECT MOTIVATION

India possesses over 7 million traditional rural artisans practicing centuries-old craft traditions including Bankura Terracotta, Assam River Bamboo, Madurai Golden Jute, Chanderi Handloom, and 4,000-year-old Lost-Wax Dokra Brass Smithing. Despite immense global demand for sustainable handmade decor and lifestyle products, rural artisans receive less than 15-20% of the end-consumer retail value due to multi-layered middlemen and severe trade entry barriers.

RuralCraft Global provides a single, unified digital platform where artisans showcase their craft heritage, and international customers purchase directly in their native currency (USD, EUR, GBP, AED, CAD, AUD, JPY). Simultaneously, the platform automates complex international trade compliance by classifying goods with 6-digit Harmonized System (HS) customs codes and automatically generating export commercial invoices with certified packing lists.

2. PROBLEM STATEMENT

Challenge Area	Real-World Problem & Impact
Intermediary Exploitation	Artisans sell to local traders at raw cost; retail margins are absorbed by intermediaries.
Customs & Tariff Complexity	Cross-border exports require 6-digit HS tariff codes, IEC (Import Export Code), and GST declarations which rural artisans cannot navigate.
Currency & Pricing Friction	International buyers require instant pricing in foreign currencies with dynamic exchange rates.
Fragmented Portals	Managing separate customer apps and admin backends causes authentication desynchronization, double deployments, and increased maintenance overhead.

3. PROJECT OBJECTIVES

1. Unified Architecture: Single deployment on Vercel handling Customer Marketplace, Artisan Studio, and Admin Command Center with zero sub-domain or multiple repo fragmentation.
2. Automated HS Code Suggestion: Algorithmic classification engine that analyzes product titles, materials, and descriptions to recommend correct WCO Harmonized Tariff Codes.
3. Instant Customs PDF Generator: Automated creation of compliant export invoices featuring exporter IEC, GSTIN, buyer declarations, and tariff notes using client-side jsPDF rendering.
4. Live Multi-Currency Conversion: Dynamic conversion between INR base prices and major global currencies (USD, EUR, GBP, AED, CAD, AUD, SGD, JPY).
5. Optimistic UI Performance: Instantaneous sub-100ms response times for product additions, edits, and deletions without blocking full-page reload spinners.

4. SYSTEM ARCHITECTURE

The platform adopts a modern 3-Tier Serverless Single Page Application (SPA) architecture designed for high scalability, zero-cold-start latency, and instant global distribution:

Architectural Layer	Technology & Responsibility
Presentation Tier (Frontend)	React 19 + TypeScript + Vite + Tailwind CSS + Framer Motion. Single bundle deployed to Vercel Edge CDN with client-side SPA routing (`vercel.json` rewrites).
Business Logic & Services Tier	Modular client-side services: `ApiService` (Supabase integration), `pdfGenerator` (customs invoices), `AuthContext` (role-based access), `CurrencyContext` (live exchange rates).
Data & Storage Tier (Supabase)	Managed PostgreSQL database with Row-Level Security (RLS), Auth trigger listeners, and relational integrity across users, products, orders, and customs records.

5. DATABASE SCHEMA DESIGN (SUPABASE POSTGRESQL)

Table Name	Attributes & Schema Structure
profiles	id (UUID), full_name, phone, role (customer/artisan/admin), designation, department, city, country, iec_code, gstin, bio, avatar_url
artisans	id (UUID), user_id, brand_name, craft_type, story, location, village, state, country, verified, rating, total_sales, phone, iec_code
categories	id (UUID), name, slug, icon, description, suggested_hs_code
products	id (UUID), artisan_id, category_id, name, slug, description, price_inr, stock, weight_kg, dimensions, material, country_of_origin, suggested_hs_code, hs_confidence, hs_code_verified, status
product_images	id (UUID), product_id, image_url, is_primary, sort_order
orders	id (UUID), order_number, user_id, customer_name, customer_email, customer_phone, total_inr, selected_currency, exchange_rate, converted_total, shipping_address (JSONB), status, tracking_number
order_items	id (UUID), order_id, product_id, product_name, artisan_id, quantity, price_inr, converted_price, suggested_hs_code, weight_kg, image_url
hs_codes	id (UUID), category, keywords (text[]), hs_code, description, country_rules, duty_notes, confidence_rating
customs_documents	id (UUID), order_id, order_number, invoice_number, invoice_date, exporter (JSONB), buyer (JSONB), items (JSONB), currency, exchange_rate, total_value_inr, total_weight_kg, verification_status

6. KEY SYSTEM MODULES & FUNCTIONALITY

System Module	Features & Technical Implementation
Module 1: Customer Storefront	Interactive explore catalog with filters for category, price ranges, and artisan regions. Responsive 3-column header navigation, instant search modal, dynamic currency switcher, animated shopping cart drawer, and multi-step international checkout.
Module 2: Artisan Studio	Dedicated portal for verified artisans to showcase cultural craft stories, upload authentic handmade items, specify craft dimensions and materials, and review direct sales analytics.
Module 3: HS Code Customs Engine	Automated trade classification algorithm matching product materials and categories against the World Customs Organization (WCO) tariff schedule. Auto-assigns codes like 4602.11 (Bamboo), 6912.00 (Terracotta), 5208.52 (Handloom Cotton), and 8306.29 (Dokra Brass).
Module 4: Commercial Invoice PDF	Client-side vector PDF generation creating official export documents with Exporter IEC & GSTIN credentials, buyer consignee address, customs tariff declarations, duty concessions under GSP/CEPA treaties, and authorized signatory blocks.
Module 5: Admin Command Center	Unified portal with tabs for Real-time Analytics, Admin Trade Profile, Products Catalog (with instant full-field Edit/Delete/Upload), Global Orders (tracking & status updates), HS Codes Master Repository, and Customs Verification.

7. HARMONIZED SYSTEM (HS) CODE REFERENCE CATALOG

HS Code	Category	Keywords / Scope	Confidence	International Customs Rule
4602.11	**Bamboo & Cane Crafts**	River bamboo baskets, rattan trays, reed utilities	**High**	Phytosanitary certificate recommended; GSP 0-2% preferential duty
6912.00	**Terracotta & Pottery**	Wheel-thrown earthen pots, clay flower pots, lamps	**High**	FDA non-toxic food-grade declaration; EPCH handmade tariff concession
5208.52	**Handloom Textiles**	Organic cotton dupattas, chanderi runners, khadi fabrics	**High**	Fiber content & wash-care label mandatory; Rules of Origin cert.
4419.90	**Teak & Woodcraft**	Hand-carved trays, sheesham bowls, botanical sculpture	**High**	CITES / Vriksh sustainable plantation timber declaration required
4202.22	**Golden Jute Carry Goods**	Biodegradable jute totes, embroidered kantha bags	**High**	Natural eco-fiber fast-track clearance across EU & GCC
8306.29	**Lost-Wax Dokra Brass**	Tribal brass figurines, bell metal cast ornaments	**High**	Non-antique contemporary handicraft declaration for ASI clearance

8. COMPREHENSIVE TECHNOLOGY STACK

Component Layer	Selected Technology & Rationale
Frontend Framework	React 19 (Latest release) with TypeScript for static type safety
Build Tool & Bundler	Vite 8.3 with @tailwindcss/vite plugin for sub-second HMR
Styling & UI Design	Tailwind CSS 4, Custom Design System (#143826 Deep Forest, #DDA15E Gold)
Micro-Animations	Framer Motion 14 (Smooth slide-in, modal scale, cart drawer transitions)
Icons & Visual Assets	Lucide React (Modern, crisp vector iconography across all views)
Analytics & Visualizations	Recharts (Responsive bar charts, monthly revenue trends, category share)
Database & Cloud Auth	Supabase Cloud PostgreSQL with row-level security and JWT sessions
Client-Side PDF Engine	jsPDF 4.2 + jsPDF-AutoTable for instantaneous vector commercial invoices
Cloud Hosting & CI/CD	Vercel Edge Network (Single unified deployment with client-side rewrites)

9. PERFORMANCE & TESTING VERIFICATION

Evaluation Parameter	Observed Benchmark & Result
Production Build Execution	**PASSED (`tsc -b && vite build` built in 1.74s, 3,203 modules transformed)**
Optimistic UI Latency	**< 50ms instant feedback on Product Edit, Delete, and Upload**
Vercel Deployment Verification	**PASSED (Single SPA routing configured in `vercel.json` with zero 404s)**
Client-Side Customs PDF Export	**PASSED (Tested via `scripts/test_pdf.ts`, renders in ~120ms)**
Multi-Currency Math Precision	**100% accurate across INR, USD, EUR, GBP, AED, CAD, AUD, JPY**

10. CONCLUSION & FUTURE SCOPE

• Project Impact: RuralCraft Global proves that technological automation in cross-border customs paperwork and direct e-commerce can eliminate predatory middlemen and preserve indigenous rural handicraft heritage.
• Academic Contribution: Demonstrates modern full-stack web engineering with React 19, serverless cloud databases (Supabase), optimistic UI architecture, automated document generation, and single-deployment devops.
• Future Roadmap: Integration of Computer Vision (AI) for automatic image-based HS code recognition, direct integration with DHL/FedEx shipping APIs for automatic airway bill (AWB) generation, and multi-lingual voice search for rural artisans.